ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Arithmetic Progressions** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 0	Common difference is -1; $a_{p+q}=0$
2	**(d)** 54	$a_n=\frac{2S_n}{n}-a$
3	**(b)** 46	$a_n=S_n-S_{n-1}$
4	**(c)** 4	$d=2p$
5	**(d)** 1325	Find n first
6	**(a)** 11	$S_n=0\Rightarrow n=1-\frac{2a}{d}$
7	**(d)** 0	Common difference is -1; $a_{p+q}=0$
8	**(b)** 59	$a_n=\frac{2S_n}{n}-a$
9	**(c)** 54	$a_n=S_n-S_{n-1}$
10	**(c)** 6	$d=2p$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$a+3d=9$, $a+8d=4$ (1); $d=-1$, $a=12$ so $a_{13}=0$ (1).	1+1
12	$a_1=S_1=7$ (1); $S_2=20$, $a_2=13$, $d=6$ (1).	1+1
13	$2a+8d=34$, $2a+9d=37$ (1); $d=3$, $a=5$ (1).	1+1
14	$a+4d=12$, $a+11d=5$ (1); $d=-1$, $a=16$ so $a_{17}=0$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\frac{6}{2}(2a+5d)=75$ and $\frac{10}{2}(2a+9d)=185$ (1); simplify to linear equations in a,d (1); $a=5$, $d=3$, AP $5,8,11,14,\ldots$ (1).	1+1+1
16	$d=4$, $a=6$ (2); $S_{16}=576$ (1).	2+1
17	Terms form an AP with $d=7$ (1); count and endpoints (1); sum $=2916$ (1).	1+1+1
	OR: $\frac{5}{2}(2a+4d)=50$ and $\frac{9}{2}(2a+8d)=162$ (1); simplify to linear equations in a,d (1); $a=2$, $d=4$, AP $2,6,10,14,\ldots$ (1).	1+1+1
18	$d=5$, $a=9$ (2); $S_{20}=1130$ (1).	2+1
	OR: Terms form an AP with $d=9$ (1); count and endpoints (1); sum $=4867$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	AP with $a=100$, $l=40$, $n=21$ (1); $S_n=\frac{21}{2}(100+40)=1470$ cm (2); middle term $=\frac{100+40}{2}=70$ cm (2).	1+2+2
	OR: AP: $a=2500$, $d=150$, $n=18$ (1); last $=a+(n-1)d=5050$ (2); total $=S_n=\frac{18}{2}(2500+5050)=67950$ (2).	1+2+2
20	AP with $a=90$, $l=30$, $n=31$ (1); $S_n=\frac{31}{2}(90+30)=1860$ cm (2); middle term $=\frac{90+30}{2}=60$ cm (2).	1+2+2
	OR: AP: $a=600$, $d=45$, $n=40$ (1); last $=a+(n-1)d=2355$ (2); total $=S_n=\frac{40}{2}(600+2355)=59100$ (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Arithmetic Progressions** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 1116	Find n first
2	**(b)** 13	$S_n = 0 \Rightarrow n = 1 - \frac{2a}{d}$
3	**(a)** 0	Common difference is -1; $a_{p+q} = 0$
4	**(a)** 52	$a_n = \frac{2S_n}{n} - a$
5	**(c)** 91	$a_n = S_n - S_{n-1}$
6	**(d)** 8	$d = 2p$
7	**(b)** 1065	Find n first
8	**(b)** 13	$S_n = 0 \Rightarrow n = 1 - \frac{2a}{d}$
9	**(b)** 0	Common difference is -1; $a_{p+q} = 0$
10	**(c)** 85	$a_n = \frac{2S_n}{n} - a$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$a_1 = S_1 = 7$ (1); $S_2 = 18$, $a_2 = 11$, $d = 4$ (1).	1+1
12	$2a + 8d = 36$, $2a + 9d = 40$ (1); $d = 4$, $a = 2$ (1).	1+1
13	$a + 5d = 11$, $a + 10d = 6$ (1); $d = -1$, $a = 16$ so $a_{17} = 0$ (1).	1+1
14	$a_1 = S_1 = 5$ (1); $S_2 = 18$, $a_2 = 13$, $d = 8$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\frac{7}{2}(2a + 6d) = 98$ and $\frac{12}{2}(2a + 11d) = 228$ (1); simplify to linear equations in a, d (1); $a = 8$, $d = 2$, AP $8, 10, 12, 14, \ldots$ (1).	1+1+1
16	$d = 3$, $a = 12$ (2); $S_{14} = 441$ (1).	2+1
17	Terms form an AP with $d = 6$ (1); count and endpoints (1); sum $= 1824$ (1).	1+1+1
	OR: $\frac{4}{2}(2a + 3d) = 42$ and $\frac{8}{2}(2a + 7d) = 164$ (1); simplify to linear equations in a, d (1); $a = 3$, $d = 5$, AP $3, 8, 13, 18, \ldots$ (1).	1+1+1
18	$d = 7$, $a = 4$ (2); $S_{18} = 1143$ (1).	2+1
	OR: Terms form an AP with $d = 11$ (1); count and endpoints (1); sum $= 7062$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	AP with $a = 70$, $l = 22$, $n = 25$ (1); $S_n = \frac{25}{2}(70 + 22) = 1150$ cm (2); middle term $= \frac{70+22}{2} = 46$ cm (2).	1+2+2
	OR: $S_n = \frac{8}{2}[2a - 140] = 880$ (2); $a = 180$ (2); last prize ₹40 (1).	2+2+1
20	AP with $a = 120$, $l = 48$, $n = 19$ (1); $S_n = \frac{19}{2}(120 + 48) = 1596$ cm (2); middle term $= \frac{120+48}{2} = 84$ cm (2).	1+2+2
	OR: $S_n = \frac{10}{2}[2a - 270] = 1150$ (2); $a = 250$ (2); last prize ₹-20 (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Arithmetic Progressions** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 63	$a_n = S_n - S_{n-1}$
2	**(a)** 10	$d = 2p$
3	**(c)** 1344	Find n first
4	**(b)** 13	$S_n = 0 \Rightarrow n = 1 - \frac{2a}{d}$
5	**(c)** 0	Common difference is -1; $a_{p+q} = 0$
6	**(c)** 44	$a_n = \frac{2S_n}{n} - a$
7	**(c)** 56	$a_n = S_n - S_{n-1}$
8	**(a)** 2	$d = 2p$
9	**(d)** 1104	Find n first
10	**(b)** 15	$S_n = 0 \Rightarrow n = 1 - \frac{2a}{d}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$2a + 8d = 32$, $2a + 9d = 34$ (1); $d = 2$, $a = 8$ (1).	1+1
12	$a + 2d = 14$, $a + 13d = 3$ (1); $d = -1$, $a = 16$ so $a_{17} = 0$ (1).	1+1
13	$a_1 = S_1 = 8$ (1); $S_2 = 26$, $a_2 = 18$, $d = 10$ (1).	1+1
14	$2a + 8d = 50$, $2a + 9d = 56$ (1); $d = 6$, $a = 1$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\frac{6}{2}(2a + 5d) = 105$ and $\frac{11}{2}(2a + 10d) = 275$ (1); simplify to linear equations in a, d (1); $a = 10$, $d = 3$, AP $10, 13, 16, 19, \ldots$ (1).	1+1+1
16	$d = 2$, $a = 15$ (2); $S_{22} = 792$ (1).	2+1
17	Terms form an AP with $d = 8$ (1); count and endpoints (1); sum $= 870$ (1).	1+1+1
	OR: $\frac{5}{2}(2a + 4d) = 75$ and $\frac{10}{2}(2a + 9d) = 325$ (1); simplify to linear equations in a, d (1); $a = 1$, $d = 7$, AP $1, 8, 15, 22, \ldots$ (1).	1+1+1
18	$d = 6$, $a = 7$ (2); $S_{12} = 480$ (1).	2+1
	OR: Terms form an AP with $d = 13$ (1); count and endpoints (1); sum $= 9265$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	AP: $a = 1500$, $d = 100$, $n = 24$ (1); last $= a + (n-1)d = 3800$ (2); total $= S_n = \frac{24}{2}(1500 + 3800) = 63600$ (2).	1+2+2
	OR: $S_n = \frac{6}{2}[2a - 250] = 1650$ (2); $a = 400$ (2); last prize ₹150 (1).	2+2+1
20	AP: $a = 800$, $d = 60$, $n = 30$ (1); last $= a + (n-1)d = 2540$ (2); total $= S_n = \frac{30}{2}(800 + 2540) = 50100$ (2).	1+2+2
	OR: $S_n = \frac{12}{2}[2a - 165] = 1410$ (2); $a = 200$ (2); last prize ₹35 (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.